

Erik Mortensen

DETAILS

Contact

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Citizenships

USA, Switzerland, France

TECHNICAL SKILLS

Programming

Python, C++, MATLAB,
LabVIEW, Simulink

Simulation

OpenSim, RobotStudio,
Star CCM+

Computer Aided Design

SolidWorks, Fusion360, Catia

Prototyping

3D Printing (FDM and SLA),
Machining, Soldering, Motor
Control, Sensor Integration

LANGUAGES

French	C2
English	C2
German	B2
Italian	B1

HOBBIES

Cooking	Traveling
Photography	Running
Homebrewing	Music

PROFILE

Newly graduated Robotics Engineer from Switzerland, seeking a first full-time opportunity in Boston. I am passionate about mechanical design and human-robot interactions. I seek innovative and elegant solutions.

EDUCATION

MSc in Robotics

Sep 2022 – Oct 2025 EPFL, Lausanne 

- Specialization: Mobile Robotics
- Including a 6-month MSc Thesis at MIT, Boston, USA 

BSc in Mechanical Engineering

Sep 2018 – Jul 2022 EPFL, Lausanne 

- Including 1 year exchange at Chalmers University of Technology, Gothenburg, Sweden 

WORK EXPERIENCE

Research Specialist at MIT AURA Lab

Feb 2026 – Present MIT, Boston 

- Pursuing research on aerial-aquatic flapping-wing drones, as part of my MSc Thesis and then continued as a Research Specialist.

Mechanical Engineering Internship at SharkNinja

Jul 2024 – Dec 2024 Boston 

- Full time intern on the R&D team of Shark's Hydro Vacuum Cleaners.
- Responsible for optimizing multiphase flow separation technology on the new [Shark StainForce](#), through design, prototyping and testing. Managing a small team of test engineers.

Sales and Production assistant at Docteur Gab's Brewery

Feb 2024 – Jul 2024 Lausanne 

- Part-time brand ambassador and labor on the production line.

Event Manager at ESN EPFL (volunteer)

Sep 2022 – Jul 2024 EPFL, Lausanne 

- Board member, in charge of the creation, organization, accounting and communication of around twenty events out of the hundred organized every year by the [association](#).

ACADEMIC PROJECTS

Pitch Control of Flapping-Wings for Efficient Hybrid Locomotion

Mar 2025 – Present

MSc Thesis at the [AURA Lab](#), MIT 

This work aims to improve the energetic efficiency of aerial-aquatic flapping-wing drones by introducing wing pitch as a new degree of freedom. After designing, building and programming a fully integrated one wing prototype, I optimized the wing motion to optimize its locomotion efficiency in the air and underwater. The results will soon be published in a scientific paper.

Mechanical Design – 3D printing – Machining – Soldering – Python – C++ – Fluid Mechanics – Field Oriented Control – Management

Robot Competition

Feb 2024 – Jun 2024

Semester Project at the [Biorobotics Laboratory](#), EPFL 

In this interdisciplinary project, six teams of three students are tasked with the design, construction and programming of an autonomous cleaning robot. The robot should be able to detect and gather plastic bricks in a 64 m² arena that features obstacles and various terrains, before bringing them back to a specific location. The project concludes with a [competition](#) between the teams. I oversaw the mechanical and electrical design and integration of the robot. I also contributed significantly to programming the state machine and obstacle avoidance. Our robot won second place, with the audience award.

Mechanical Design – 3D printing – Laser Cutting – Arduino – Python – Teamwork

Tensegrity-based Robot Leg Design with Variable Stiffness

Sep 2023 – Jan 2024

Semester Project at [Laboratory of Intelligent Systems](#), EPFL 

During this project, I designed, built and tested a robotic leg to be used for robot locomotion. The resulting prototype is a tensegrity based fully actuated robotic limb, with a bio-inspired mechanism to actively vary the compliance of its joints. The results have been presented at [IEEE Robosoft 2025](#) conference and published [online](#).

Mechanical Design – Bio-inspired – 3D printing – Arduino – MATLAB – Soft Robotics

Chibbun – Urban Seating for Elderly

Nov 2022 – Aug 2023

Lausanne  and Seoul 

In a team composed of engineers and industrial designers, we developed a public chair that can be easily installed in cities to promote elderly inclusivity in urban areas. This project was part of the course Design for Sustainability [HUM-398](#) at EPFL and ECAL (art and design university of Lausanne). Selected to participate to the [KimCheese](#) exchange program between four universities in Switzerland and South-Korea, we had the opportunity to present and test our prototype in Seoul during summer 2023.

Industrial Design – Sustainability – Multi-Disciplinary Teamwork

3D Object Detection using Monocular Camera

Nov 2022 – Jul 2023

EPFL 

I implemented and trained a neural network that can recognize the cars, their 3D location and their orientation in 2D images to be used for autonomous robot locomotion. This project has been realized in a group of two, as part of the Deep Learning for Autonomous Vehicles [CIVIL459](#).

Reinforcement Learning – Image Processing – Python – Teamwork

Keep Focus – Auto Focusing Magnifying Ocular

Nov 2022 – Jul 2023

[Reconfigurable Robotics Lab](#), EPFL 

In a group of five, we designed and produced a prototype of magnifying ocular that automatically adjusts the focus to the object of interest. This project, aiming to improve the working conditions of surgeons and watchmakers, has been realized in the scope of the course Mechanical Product Design and Development [ME-410](#). The concept is currently being developed as a [startup](#).

Mechanical Design – Product Development – 3D Printing – Arduino – Teamwork
